AR Geometric Pattern Studio: Lines, Circles, and Islamic Geometric Art

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Application Description

We propose a smartphone AR app for iPhone, developed on a MacBook with Unity, AR Foundation/ARKit, and Xcode. The user scans a textured tabletop and places one anchored art board. On it, they build a small design using either tools and buttons provided on the app or a fingertip cursor and pinch gestures, and the app repeats the design across the board in a square grid. This is a matter of scope and we might only go with the former as the latter is more complicated. An enlarged, editable unit cell sits beside the repeated pattern, so editing the source motif updates every copy.

The app combines two connected activities. In *Free Design*, users assemble triangle, square, and hexagon stamps, plus lines and circles, then recolor them and use a four- or eight-copy radial helper. In *Guided Eight-Pointed Star*, users follow a digital compass-and-straightedge lesson: place a guide circle, mark eight equally spaced points on it, join alternate points to form one square, join the rest to form a second square rotated by 45 degrees, and then hide the guide circle and diameters to reveal the star. Optional connector segments run from the star's cardinal tips to the cell's edge midpoints, so the repeated cells join into a connected pattern. Lines and circles are defined by control points, and snapping to endpoints, centers, circle-division points, and intersections means the hand does not need to draw perfect shapes. Every action also works with touch or mouse, and undo, delete, clear, and save/load are included.

Motivation

We are interested in this project because geometric ornamentation is a natural fit for AR and hand interaction: the same construction can be drawn on a real table, inspected from any angle, and repeated into something that looks like finished artwork. Islamic geometric art is a rich example of how circles and straight lines alone can generate elaborate designs. The Metropolitan Museum of Art's teaching materials introduce these designs through exactly these constructions, and we will use an attributed Met example for our one guided lesson. Our app makes the construction steps visible, so users can see where a star's vertices come from, hide the guides to see the artwork, and watch a single motif become a repeating pattern. We also find the technical side appealing: exact geometry, camera-based hand tracking, and AR placement all have to work together on a real device. This project will enable artists to first create an augmented pattern on a canvas, and that will allow them to draw these patterns on the canvas a lot more easily and save time. Creating construction lines on canvas is tedious and this application saves time by making it quicker to draw the construction lines as augmented reality via a phone screen.

Scope

To fit the effort of three assignments, we are keeping the scope narrow. The required features are: one board; stamps, lines, and circles; a small palette and line-width control; pinch plus touch/mouse input; snapping and a separate guide layer; the single guided eight-pointed lesson; square-grid repetition with a bounded row/column count; undo; and save/reload (the design is rebuilt on a newly placed board). The milestones are: (1) a signed iPhone build with board placement, a working hand marker, and basic line/circle tools; (2) gesture construction, snapping, and the eight-pointed lesson with connected repeats; (3) persistence, performance profiling, user testing, and demo preparation. A six-fold lesson with a triangular lattice, more star recipes, trimming, interlacing, and a standalone Mac editor are stretch goals only. We will not attempt all wallpaper groups, gapless tessellation of arbitrary shapes, or a general constraint solver.

Main Challenges and Our Plan

- **Hand tracking on the phone.** Apple's Vision framework gives image-space hand joints, not AR-world positions or pinch events, and it must share the camera with AR Foundation. We will write a small Objective-C++ bridge, copy downsampled camera frames into a buffer we own, and run one detection job at a time. We will compute pinch gestures ourselves with confidence limits, separate on/off

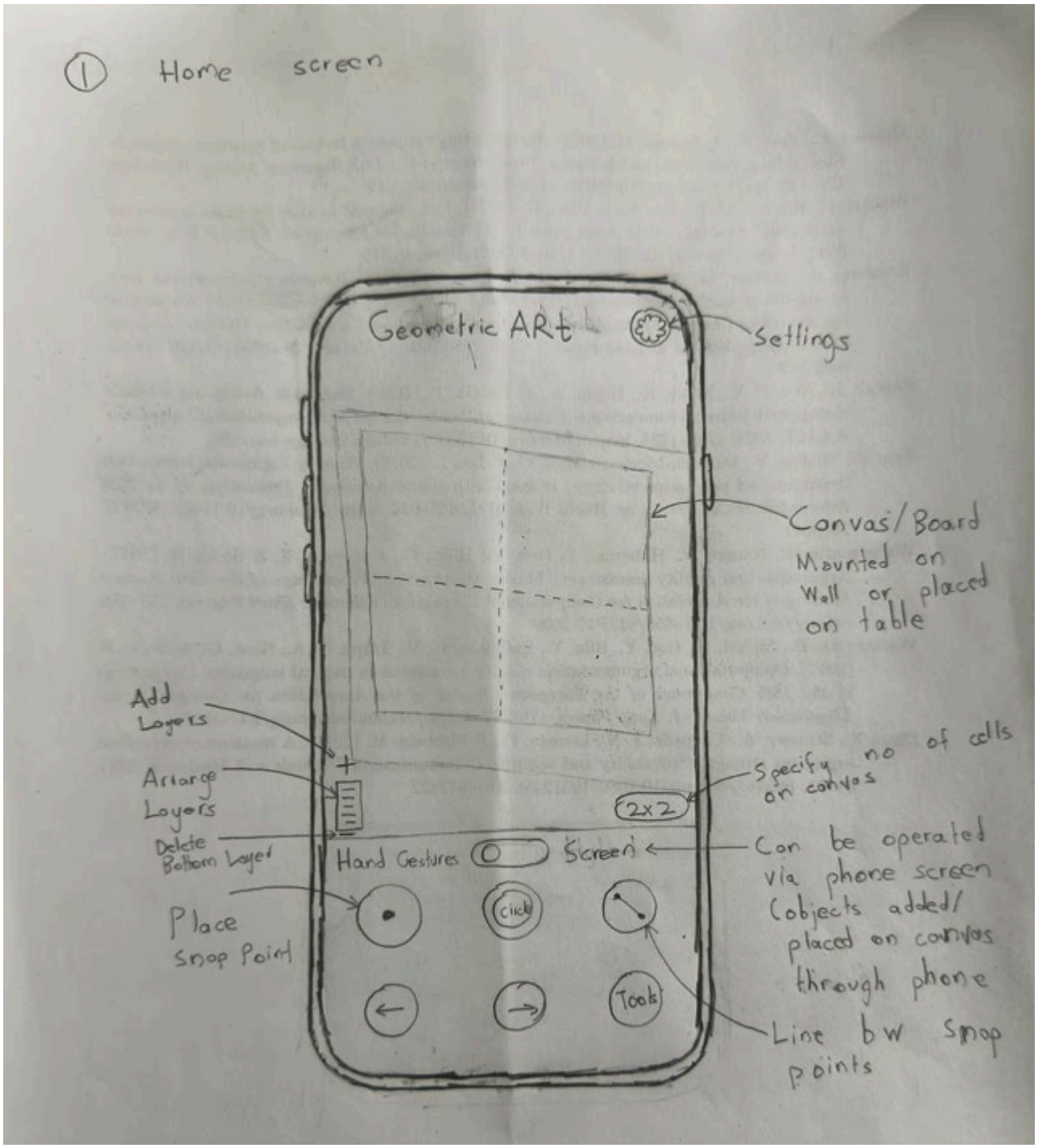
thresholds, and debouncing. We will map joints through the real image orientation and display transform and check the cursor against the fingertip at several screen positions. Stale detections are rejected using timestamps. This is the biggest risk, so we test it on the physical phone in Milestone 1.

- **Precision with a noisy hand.** Users cannot trace perfect shapes in the air, so we store exact primitives (line endpoints, circle center/radius) placed by snapping, work in an enlarged cell, and give touch alternatives for every command. If tracking is lost mid-action, the unfinished shape is cancelled, and editing resumes only after a fresh pinch.
- **Geometry edge cases.** Tangent circles, parallel or coincident lines, and zero-length shapes can break intersection snapping. We will keep the geometry in plain C# in the board's 2D space, enforce minimum lengths and radii, and unit-test these cases on the Mac without starting AR.
- **Connected repetition.** Repeating a motif does not guarantee that lines meet at cell edges. We will claim seamless joins only for the guided star, whose connectors have known matching boundary positions, and show neighboring cells while editing so users can see the seams. Free designs may have gaps or overlaps, and we will say so. The guided lesson uses a small linked center/radius parameter graph rather than a general solver.
- **Performance and device setup.** Many repeated copies could slow rendering. We will store only source geometry, cap rows and columns, reuse geometry, and profile on the phone, with a target of at least 30 FPS. Signing and provisioning will be set up early, since free developer builds expire after seven days.

Evaluation

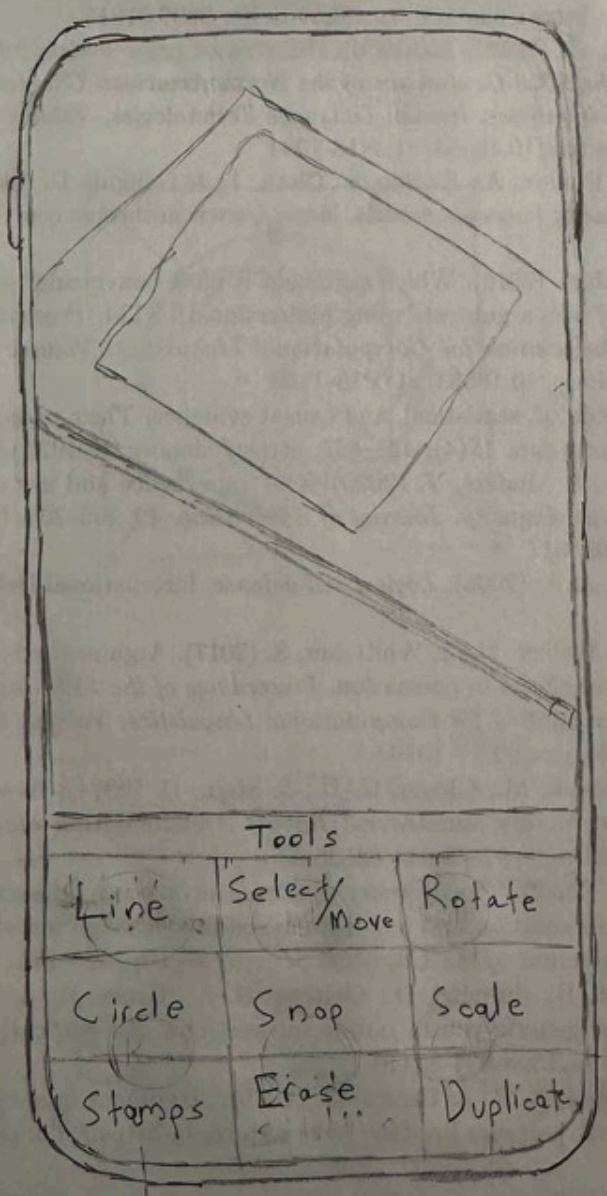
We will have three to five users build a stamped motif and the eight-pointed star, hide the guides, repeat, edit, undo, and save/reload, recording task completion, time, false gestures, and tracking loss. Our prototype targets are 8 of 10 successful placement/drag trials, a median of at least 30 FPS over two minutes, and guided seam positions matching within 0.00001 times the cell side. These are targets, not results.

Sketches

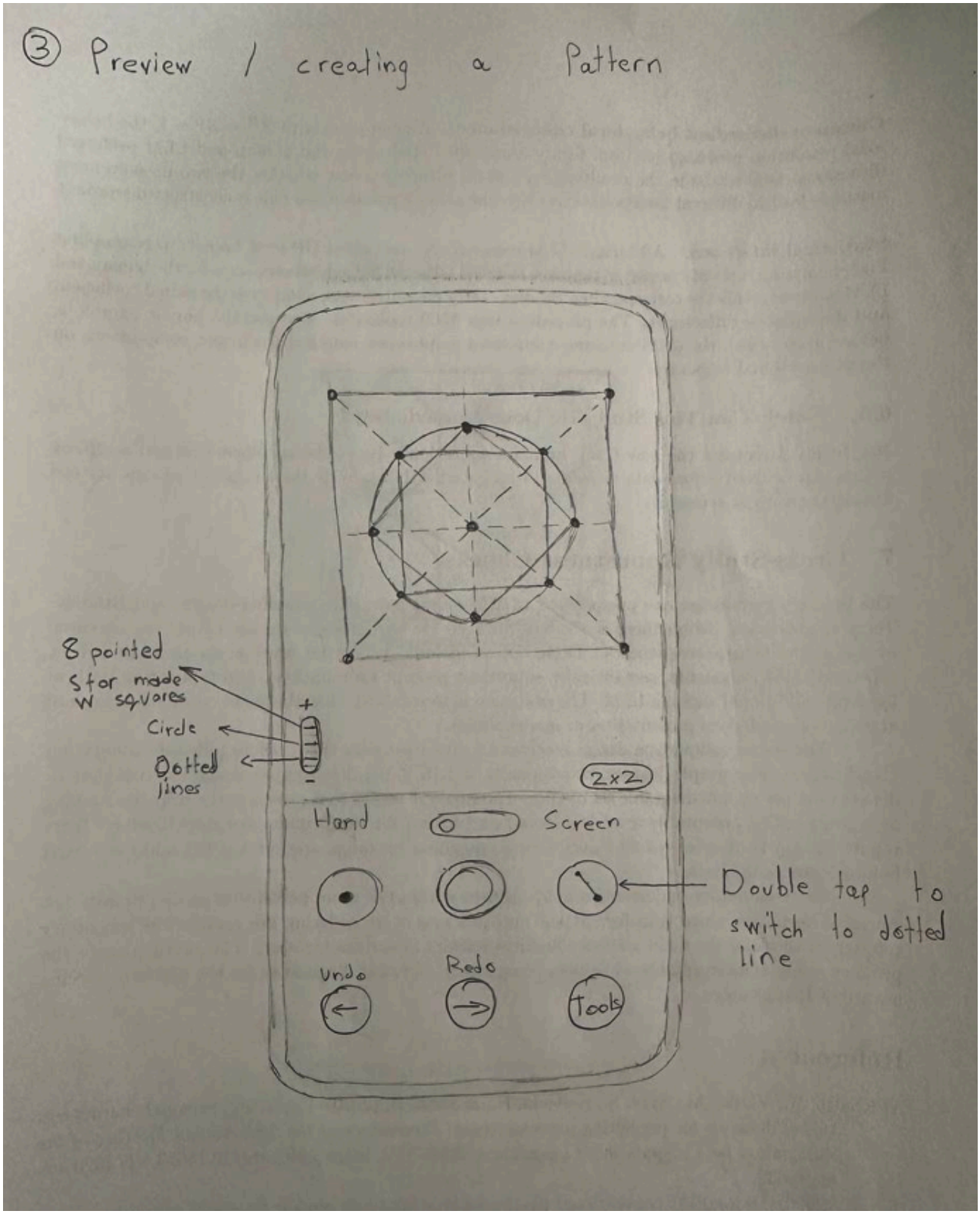


1. **User opens the application and mounts the board, sees the quick menubar**

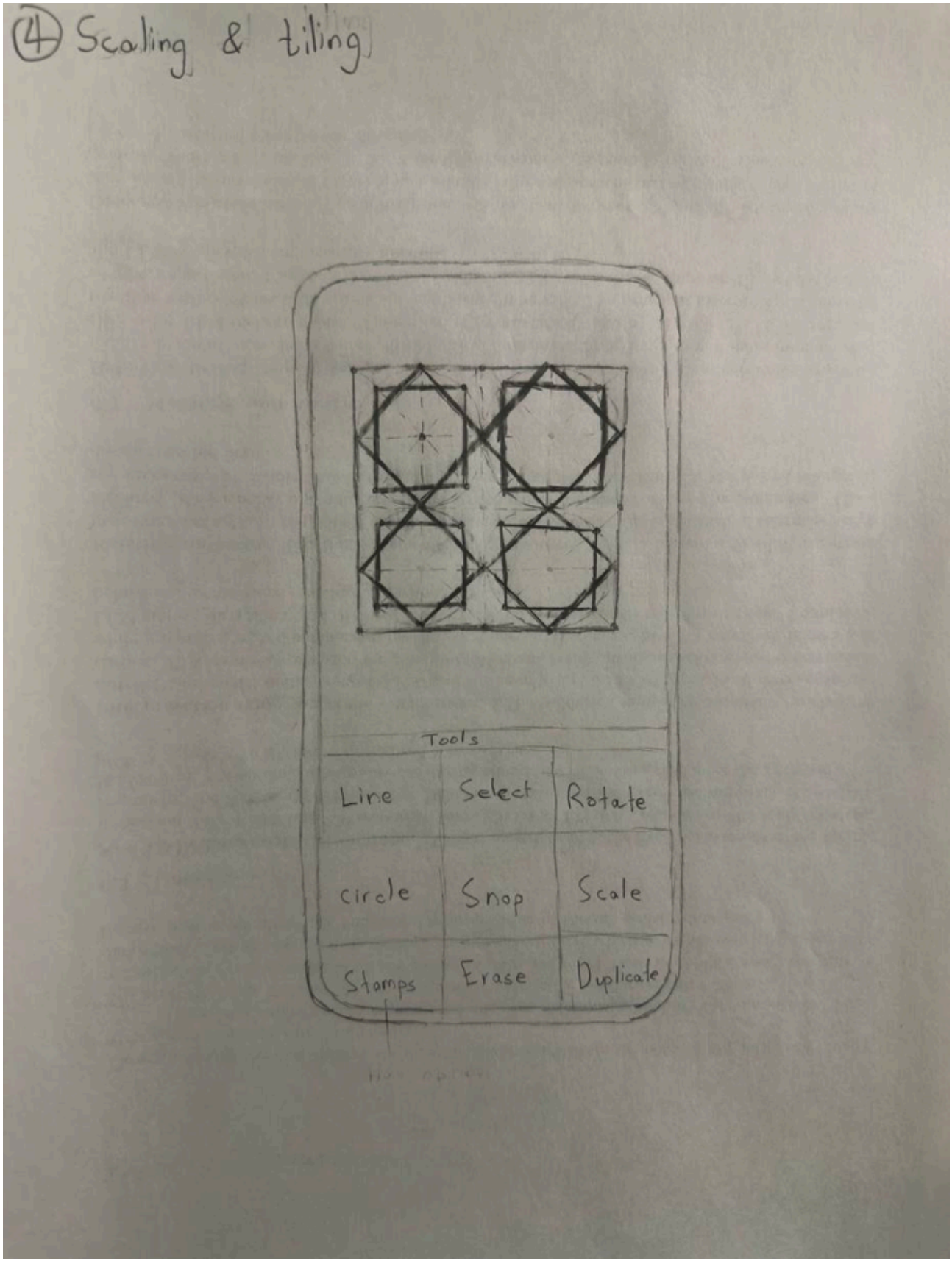
② Tools Screen



2. **User checks out the tools available**



3. **User makes a motif, uses layers to make it easy to delete unnecessary lines.**



4. **User switches to tools mode, scales and tiles the pattern on a 2*2 grid using duplicate, select and then dragging the shape. Snap points are essential to snap the tiles into place.**

References

Computer Generated Islamic Star Patterns (Kaplan, 2000)

Summary: This paper explains how a computer can construct Islamic star patterns from smaller geometric designs. It begins with a tiling–shapes arranged to cover a surface–and places stars or flower-like designs called *rosettes* inside suitable polygons. Lines are then extended into the remaining spaces to connect the designs. Removing the underlying construction grid reveals the decorative pattern. The paper also presents different rendering styles, including outlines and interwoven-looking bands, through the interactive program Taprats.

How it relates to our proposal: Our application similarly separates creating a small design, or *motif*, from repeating it into a larger pattern. Users will edit an enlarged square cell and see its copies update across the AR board. The paper's separation of construction geometry from finished artwork also relates to our option to show or hide guide circles and lines.

How our project differs: Kaplan focuses on generating patterns from mathematical rules and different tilings. Our project focuses on letting users construct designs themselves through simple tools and a guided lesson. We will use square-grid repetition rather than implement the paper's broader automatic pattern-generation system.

Islamic Star Patterns in Absolute Geometry (Kaplan & Salesin, 2004)

Summary: Kaplan and Salesin introduce Najm, a system that generates Islamic star patterns in different geometric settings. It supports patterns on flat surfaces, spheres, and the hyperbolic plane, where geometric relationships differ from ordinary flat geometry. The system combines polygonal tilings, adjustable star and rosette designs, and an algorithm that fills remaining regions. Its main contribution is a mathematical framework that can produce related patterns across these different settings.

How it relates to our proposal: The paper offers a useful example of defining designs through adjustable geometric values. Our guided star will also be constructed from related values: its center, radius, and equally spaced points. This approach allows the application to rebuild the design consistently when a controlling value changes, instead of requiring users to reposition every line individually.

How our project differs: Our application will use ordinary two-dimensional geometry on a flat AR board. We will support a small number of understandable constructions rather than reproduce Najm's extensive mathematical capabilities or general region-filling algorithm.

Islamic Star Patterns from Polygons in Contact (Kaplan, 2005)

Summary: This paper develops a computerized version of Hankin's "polygons-in-contact" method. Starting with a tiling, pairs of lines extend inward from the midpoints of polygon edges. The lines meet other lines to form decorative motifs, while neighboring motifs connect at shared boundaries. Changing the angle of the lines produces different patterns. The paper also explores variations that move their starting positions or gradually change the design across a surface.

How it relates to our proposal: This work is especially relevant to making repeated designs connect. Copying a motif into neighboring cells does not automatically make their lines meet. Our guided eight-pointed star addresses this by using optional connector segments that reach matching positions at the cell boundaries. Showing neighboring cells during editing will help users inspect those connections.

How our project differs: Our lesson constructs a star using a circle and two overlapping squares, rather than implementing Hankin's complete method. It allows the users to get an intuition of how to make these patterns instead of making it an automatic process. We will provide known matching connectors for this specific construction. We will not automatically calculate compatible line networks for every shape users create in Free Design mode.

MediaPipe Hands: On-device Real-time Hand Tracking (Zhang et al,, 2020)

Summary: This paper presents a mobile hand-tracking system that uses an ordinary color camera. One model finds the palm, and another estimates 21 hand landmarks, including finger joints and fingertips. During tracking, information from previous frames helps locate the hand without repeating the full detection process every time. The paper demonstrates how these landmarks can support gesture recognition and AR effects.

How it relates to our proposal: A similar separation is useful in our interaction design: first locate the hand, then interpret its movement as an action. Fingertip positions could control a cursor, while the relationship between the thumb and index finger could help identify a pinch. Our application would then map that input onto the drawing board.

How our project differs: Our iOS proposal uses Apple Vision rather than MediaPipe. We will use an existing hand detector instead of developing or training a new one. Our contribution is the drawing interaction built around its output. MediaPipe's results do not establish Apple Vision's architecture or guarantee our application's performance.

Citations

Kaplan, C. S. (2000). Computer generated Islamic star patterns. In R. Sarhangi (Ed.), *Bridges: Mathematical connections in art, music, and science* (pp. 105–112). Bridges Conference. https://archive.bridgesmathart.org/2000/bridges2000-105.html

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